进度汇报

2020.5.15

胡敬玉

目录

- Kendall, A., Gal, Y., n.d. What Uncertainties Do We Need in Bayesian Deep Learning for Computer Vision? in NIPS, 2017.

- 脑肿瘤分割中的数据不确定性

不确定性分类

- Epistemic Uncertainty（模型不确定）

P(W|X, Y)

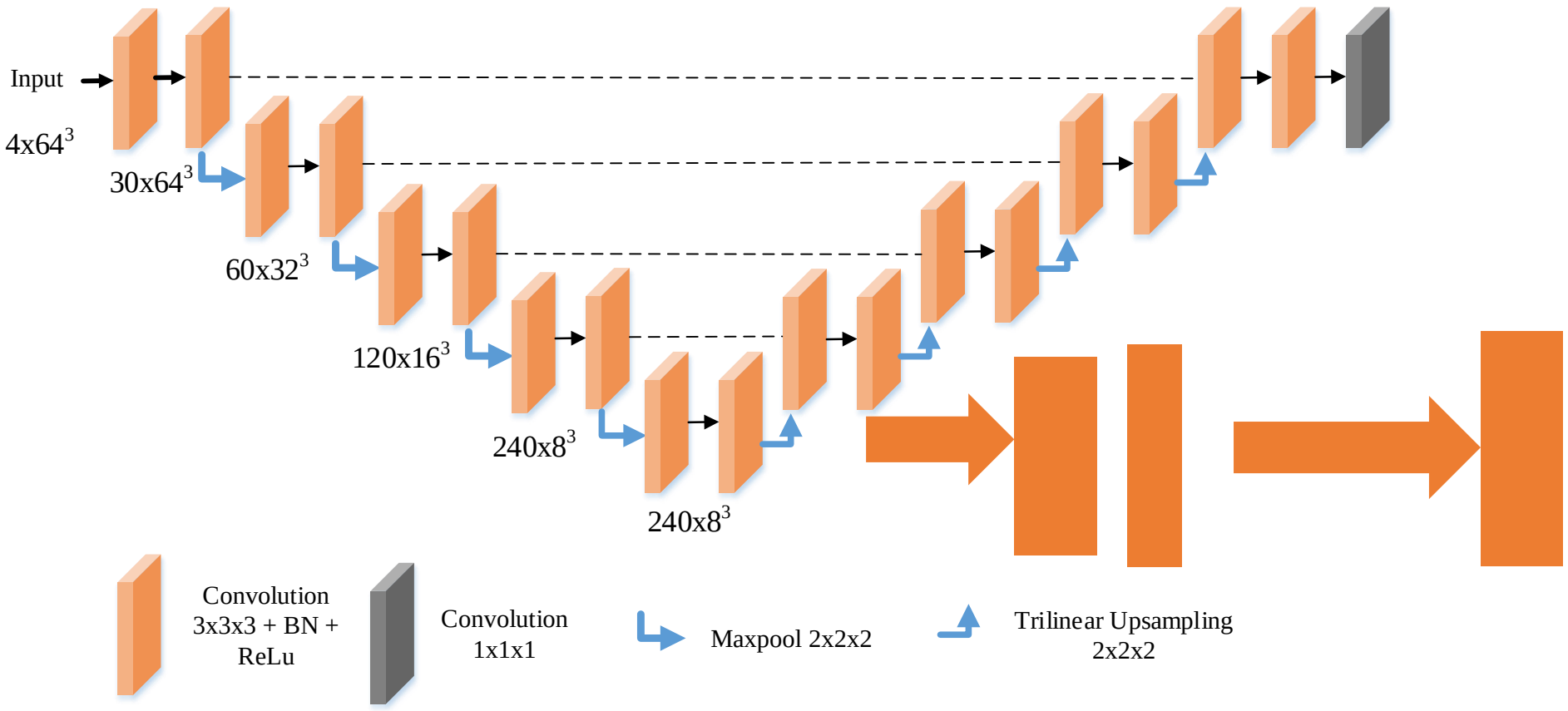
- Aleatoric Uncertainty（数据不确定）

Noise in data σ^2

$$[\hat{\mathbf{y}}, \hat{\sigma}^2] = \mathbf{f}^{\widehat{\mathbf{W}}}(\mathbf{x})$$

$$\mathcal{L}_{BNN}(\theta) = \frac{1}{D}\sum_i \frac{1}{2}\hat{\sigma}_i^{-2}||\mathbf{y}_i - \hat{\mathbf{y}}_i||^2 + \frac{1}{2}\log\hat{\sigma}_i^2$$

网络结构

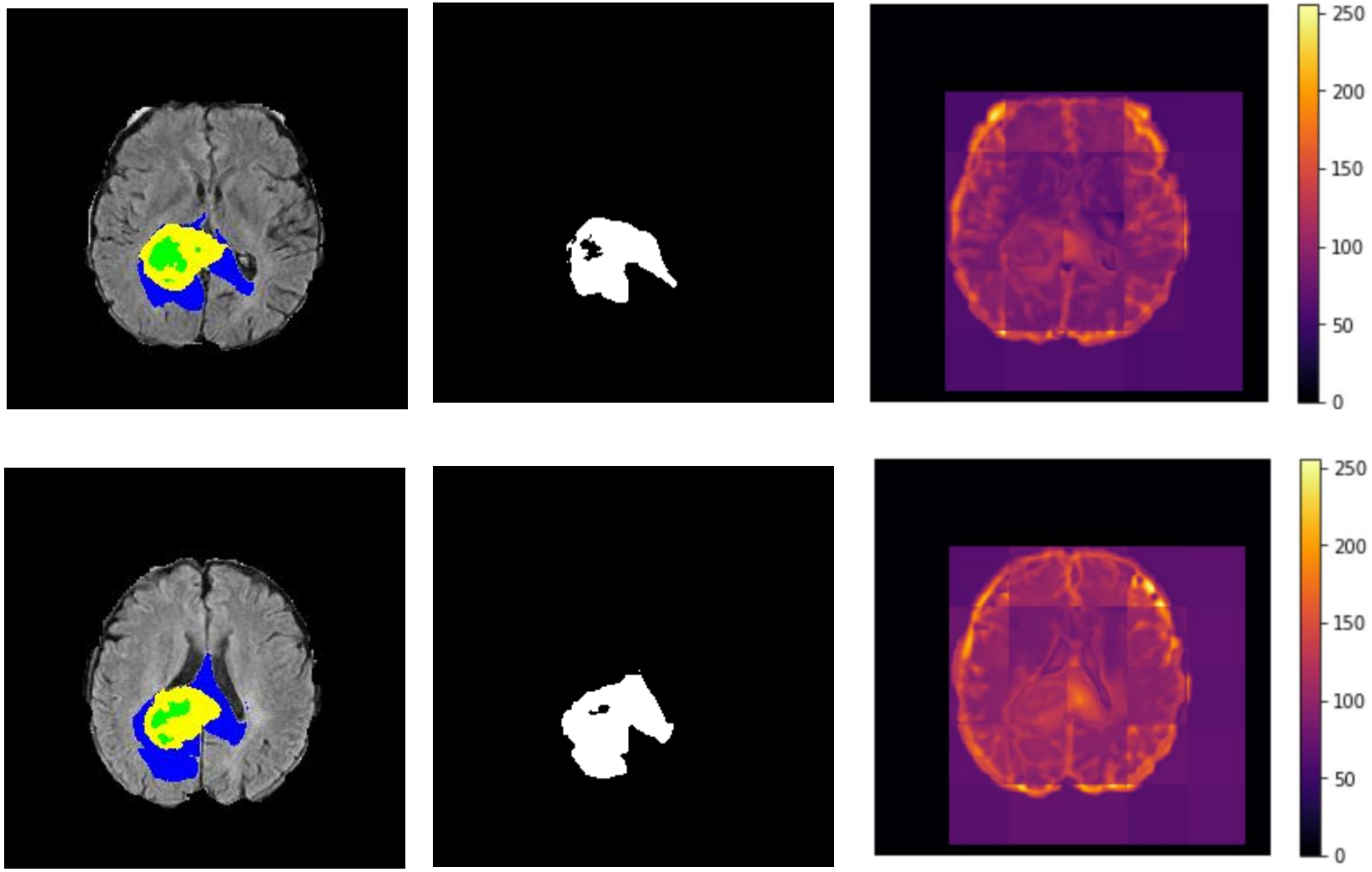


$$\mathcal{L}_{BNN}(\theta) = \frac{1}{D}\sum_i \frac{1}{2}\hat{\sigma}_i^{-2}||\mathbf{y}_i - \hat{\mathbf{y}}_i||^2 + \frac{1}{2}\log\hat{\sigma}_i^2$$

定量分析

Method	Dice Coefficient		
	Com.	Core	Enh.
Unet	85.78	78.09	73.5
Uncertainty Net	82.03	71.75	66.58

定性分析



Thank You!